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RESEARCH-ARTICLE

Energy management of networked, solar cells powered, wireless sensors

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ABSTRACT

Solar cells combined with power management algorithms enable the dynamic scheduling of Wireless Sensor Networks applications in a reference period, where the objective of the scheduling is to maximize the application quality level while conserving an energy level sufficient to constantly maintain the sensor operation. In this paper we consider networked, solar cells powered wireless sensors and we propose an algorithm aims to find a global, (sub)optimal scheduling that maximizes the overall quality of service in the sensors and keeps the system energy neutral, thus ensuring that the system works uninterruptedly.

Categories and Subject Descriptors

C.2.m [Computer-Communication Networks]: Miscellaneous;
I.6.4 [Simulation and Modeling]: Model Validation and Analysis—*Algorithms, Design, Simulation*

Keywords

Power Management; Solar Cells; Quality of Service; Wireless Sensor Networks.

1. INTRODUCTION

A Wireless Sensor Network (WSN) comprises a large number of low-power wireless embedded devices, called sensors, that can be easily deployed in a given area in order to collect environmental data. The sensors send any related sensed data to a special node (called sink) that act as a gateway between the WSN and external networks. Most of the commercial sensors available nowadays are powered by limited-capacity batteries that provide the energy required to sustain their operations. Consequently, the sensor lifetime is limited: when the battery drops below a minimum charge level, the sensor stops working. Such a limitation may be addressed by means of energy harvesting systems that use the energy obtained naturally from the environment to power on the sensors. However, since the energy production is generally not continuous, the sensors still use batteries that are recharged when there is an excess of energy

production that guarantees their operation under an insufficient production. Therefore, an efficient energy management is necessary to guarantee an infinite sensor's lifetime while maintaining the activities of the sensors. In our work we model the sensors activities as tasks with a cost and a quality level, which expresses how much the task meets the user requirements. Based on this model, we devise a scheduling strategy that aims at maximizing the quality of service (QoS) while keeping the sensor energy neutral (i.e. within a time frame the energy production equals the energy consumption). Specifically, we consider a simple scenario that comprises two networked devices, one sensor and one sink, that communicate wirelessly and that employ an energy harvesting system based on solar cells combined with embedded batteries. We propose a low-complexity algorithm aimed at finding a (sub)optimal application scheduling that concerns to both the sensor and the sink, such that the overall quality of the scheduling is maximized, the energy is preserved to guarantee the applications execution even in conditions of darkness or low production, and the tasks selected at each time for the sensor and for the sink are compatible (i.e. they can work together).

2. RELATED WORK

Adaptive sampling techniques adjust the sampling rate of a sensor based on the spatial/temporal correlation of the data [1] to save energy. These techniques are often used in sensors equipped with some energy harvesting system to estimate the energy that will be produced in a future time frame. Among the forms of energy harvested from the environment, the energy produced from the sun may be predicted with reasonable accuracy. In [4] the concept of *energy neutrality* is introduced to state that the energy consumed by a sensor in a period of time is always less or equal than the energy harvested from the environment in the same period. Based on this, they propose an algorithm that divides the sensor lifetime in time slots, and that assigns to each slot the maximum sampling rate that keeps the system energy neutral. Although QoS is not mentioned explicitly, we observe that increasing the sampling rate corresponds to rising the QoS of the application running on the sensor. QoS of the WSN applications is also related to other performance metrics such as throughput, service differentiation, coverage, exposure, reliability and, lately, energy consumption. We have addressed the QoS optimization while keeping the energy neutral condition in two previous works [2] [3]. Differently than the above works, in this paper we aim at optimizing the QoS in a context of networked sensors, where

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the optimization takes into account also the **compatibility of the scheduling plans of different devices** and the fact that energy production and consumption is, in general, not the same on all devices. To the best of our knowledge, this is the first work that consider the optimization of QoS in energy harvesting sensors under this perspective.

3. SYSTEM MODEL

We consider a scenario that comprises one sensor and one sink powered by solar cells. The efficiency of a solar cell (η) is a measure of the energy conversion efficiency of the incident light absorbed by a solar cell into electric power and is computed as $\eta = \frac{P_{max}}{D \times S}$, where P_{max} is the maximum output power provided (in Watts), D is the irradiance, defined as the density of incident power on a surface under standard assumptions (in W/m^2) and S is the surface area of the solar cell (in m^2). The output power is computed as $P_{out} = \eta \times D \times S$ and the average energy production E_p of the solar cell is P_{out}/V_{id} . To obtain the instantaneous irradiance of the solar cell at time t , we apply a correction factor depending on the current day and solar time. Thus, at time t the instantaneous irradiance can be expressed as $D(t) = \frac{(t-h)^2}{4p} + D_{month}$, where t is any hour between 0:00 and 23:00, h is the hour of maximum solar light (12:00am), D_{month} is equal to the average irradiance in a month and $p = 9/D_{month}$, where 9 is the number of standard hours of solar light. The estimated energy production at time t is $E_p(t) = \frac{D(t) \times \eta \times S}{V_{id}}$.

Our **task model** considers a sensor that executes a subset of the available tasks in $\Theta = \{T_0, \dots, T_\alpha\}$ in a given period of time. A task T_i ($i > 0$) is represented by a triple $T_i = \langle t_i, p_i, w_i \rangle$, where t_i is the execution time of the task, p_i is the period of execution of the task (note that must be $p_i > t_i$), and w_i is the energy consumption of the sensor due to the execution of task T_i . A special task T_0 (also called Idle task) is executed when there is no other task to execute, which means that T_0 does not have a period and a predefined execution time. The sensor executes a scheduling plan P that is defined as a finite sequence of tasks in Θ . Note that since the tasks are periodic, P is also cyclic. The cost of execution of P per unit of time is given by $c = \sum_{T_i \in \Theta} \frac{w_i}{p_i} + w_0 \left(1 - \sum_{T_i \in \Theta} \frac{t_i}{p_i}\right)$. Let us assume n possible scheduling plans of the sensor $P_1^N, \dots, P_n^N$ and m scheduling plans of the sink $P_1^S, \dots, P_m^S$. We use notation P_i^X , with $X \in \{S, N\}$, to denote the scheduling plan i that is either of the sensor (if $X = N$) or of the sink (if $X = S$). For a given scheduling plan P_i^X are defined its energy cost per unit of time c_i^X and a defined quality level q_i^X . Let us also assume two scheduling plans P_i^N and P_j^S with $i \leq n, j \leq m$. To enable the communication between the sensor and the sink, it exists a sending task $T_x \in P_i^N$ and a receiving task $T_y \in P_j^S$. Then, P_i^N and P_j^S are compatible iff $(p_x) \geq (p_y)$ and $(p_x + t_x) \leq (p_y + t_y)$. To express the compatibility among any pair of scheduling plans of the sensor and of the sink, we use the matrix C , composed by n rows and m columns, that represent the compatibilities by using simple binary notation: if $C(i, j) = 1$ then P_i^N and P_j^S are compatible plans; otherwise $C(i, j) = 0$, which means that they are incompatible and cannot be run together.

In our **energy model** we define B_{max}^X and B_{min}^X to be the maximum and minimum battery capacity of the device,

respectively. The energy available in the batteries may increase (or decrease) depending on the solar irradiance in a place and instant of time. As stated in [4] a sensor can operate uninterruptedly if its energy management is *energy neutral*. This means that, in a reference period of time $[t', t'']$, must be $B_{t'} = B_{t''}$. In our case, since the energy harvesting system is based on solar cells, we consider a time frame of 24 hours divided into σ slots of duration $\delta = \frac{3600 \times 24}{\sigma}$ seconds. Let $e_p^N(i)$ and $e_p^S(i)$ be the energy production at the end of slot i and $e_e^N(i)$ and $e_e^S(i)$ be the expected energy production (estimated by using the model discussed before) at the end of slot i , of the sensor and the sink, respectively. Similarly, let $c^N(i)$ and $c^S(i)$ be the energy cost due to the execution of a scheduling plan during the slot i , in the sensor and the sink, respectively. We denote with $B^X(i)$ the battery level at the end of slot i of the device X . Let also $\phi \in [0, 1]$ be the charging efficiency of the battery and the rectifier function $[x^X]^+$ be defined as follows: if the charge x tends to overflow the battery above B_{max}^X , then the battery charge will be B_{max}^X also at the end of the slot. Similarly, if the charge tends to underflow the battery below 0, then the battery charge will be 0 also at the end of the slot. Under these assumptions, and disregarding the power leakage of the battery, we express the estimated battery level of the sensor and of the sink for each slot $i \in [1, \sigma]$ as $B^X(i) = \max\{B_{max}^X, B^X(i-1) + \phi[e_e^X(i) - c^X(i)]^+ - [c^X(i) - e_e^X(i)]^+\}$, where $B^X(0)$ is the initial battery level at the beginning of slot 1, the second term accounts for the energy which is produced and not consumed by X and the last term accounts for the energy that is drained from the battery in X .

4. POWER MANAGEMENT ALGORITHM

We assume that the scheduling plans and their quality levels for the sensor and the sink are known, and that they are energy neutral at least in some condition (e.g. the scheduling plans that keep in idle the sensor and the sink are always compatible). Different scheduling plans may implement completely different functionalities with different performance (e.g. different duty cycles). Our purpose is to find a (sub)optimal assignment of plans to slots such that the overall QoS in a time frame is maximized while maintaining the overall system energy neutral. The vectors S^N and S^S contain the allocation of the scheduling plans to each time slot for the sensor and for the sink, respectively, being its energy budget $B^X(\sigma) - B^X(0)$. We define the vector q of the quality level, the vector g of compatibility of the scheduling plans, and the vectors c^X the energy costs of X . Specifically, for a given S^N and S^S and a time slot $i \in [1, \sigma]$ let $S^N[i] = j$ and $S^S[i] = k$, we have that $q[i] = \min\{q_j^N, q_k^S\}$, where q_j^N and q_k^S are the quality of the plan j and k assigned for the sensor and the sink, respectively at slot i ; $g[i] = C(j, k)$; $c^N[i] = c_j^N$, and $c^S[i] = c_k^S$. We formalize it as an optimization problem:

$$\max \sum_{i=1}^{\sigma} q[i] \wedge \prod_{i=1}^{\sigma} g[i] = 1 \quad (1)$$

$$B^N(0) \leq B^N(\sigma) \wedge B^S(0) \leq B^S(\sigma) \\ B_{min}^N \leq B^N(i) \wedge B_{min}^S \leq B^S(i) \forall i \in [1, \sigma]$$

We propose an heuristic aimed at finding an admissible solution that approximates the optimal solution. Let Φ^X be a set of scheduling plans. We define $opt(\Phi^X)$ as the schedul-

Sensor Plan	T_1	T_2	T_3	T_4	T_0	c	$q(\%)$	Efficiency
N1	$\langle 2, 5, 10 \rangle$	$\langle 1, 5, 15 \rangle$	$\langle 1, 5, 8 \rangle$	—	T_0^1	7.0	100	14.28
N2	$\langle 1, 10, 3 \rangle$	$\langle 1, 10, 5 \rangle$	$\langle 1, 10, 5 \rangle$	$\langle 1, 10, 7 \rangle$	T_0^6	3.2	90	28.12
N3	$\langle 5, 15, 10 \rangle$	—	—	—	T_0^{10}	2.0	85	42.50
N4	$\langle 2, 20, 3 \rangle$	$\langle 4, 20, 4 \rangle$	—	—	T_0^{14}	1.75	80	45.71
N5	$\langle 20, 30, 10 \rangle$	—	—	—	T_0^{10}	1.0	75	75
N6	$\langle 26, 30, 1 \rangle$	—	—	—	T_0^0	0.3	70	233.3
Sink Plan	T_1	T_2	T_3	T_4	T_0	c	$q(\%)$	Efficiency
S1	$\langle 1, 3, 2 \rangle$	$\langle 1, 3, 10 \rangle$	$\langle 1, 3, 12 \rangle$	—	T_0^0	8.0	100.0	12.5
S2	$\langle 1, 5, 10 \rangle$	—	—	—	T_0^4	3.6	98.0	27.2
S3	$\langle 1, 6, 7 \rangle$	$\langle 1, 6, 3 \rangle$	$\langle 1, 6, 5 \rangle$	$\langle 2, 6, 2 \rangle$	T_0^1	3.16	95.0	30.0
S4	$\langle 2, 10, 3 \rangle$	$\langle 6, 10, 10 \rangle$	—	—	T_0^2	1.7	90.0	52.94
S5	$\langle 7, 10, 10 \rangle$	—	—	—	T_0^3	1.6	85.0	53.13
S6	$\langle 1, 20, 2 \rangle$	$\langle 1, 20, 1 \rangle$	$\langle 1, 20, 3 \rangle$	$\langle 10, 20, 1 \rangle$	T_0^7	1.05	80.0	76.19
S7	$\langle 27, 30, 7 \rangle$	$\langle 2, 30, 10 \rangle$	—	—	T_0^0	0.63	60.0	94.73
S8	$\langle 32, 35, 1 \rangle$	—	—	—	T_0^3	0.2	50.0	250.0

Table 1: Scheduling plans for the sensor and for the sink employed in the simulation.

ing plan $P_i^X \in \Phi^X$ such that $\forall P_j^X \in \Phi^X$ holds $\frac{q_i^X}{c_i^X} \geq \frac{q_j^X}{c_j^X}$ and $q_i^X > q_j^X$ if $\frac{q_i^X}{c_i^X} = \frac{q_j^X}{c_j^X}$ (i.e. P_i^X is the most efficient scheduling plan in Φ^X). Note that a plan P_j^X for which $q_j^X < q_i^X$ and $c_j^X \geq c_i^X$ for some i is inefficient and can be excluded. We define $\Phi_q^X = \{P_i^X \in \Phi^X : q_i^X > q\}$ as the set of scheduling plans in Φ^X that have a quality level above q , and $\Phi_c^X = \{P_i^X \in \Phi^X : c_i^X < c\}$ as the set of scheduling plans in Φ^X that have a cost below c . Given P_i^N and a set of plans of the sink Φ , the subset of scheduling plans in Φ compatible with P_i^N is $\text{comp}(\Phi, P_i^N) = \{P_j^S \in \Phi : C(i, j) = 1\}$. We make the following assumptions:

ASSUMPTION 1. *Given the scheduling plans P_i^N and P_j^N , let us consider two scheduling plans for the sink P_h^S, P_k^S such that P_h^S is the minimum cost scheduling plan for the sink in the set $\text{comp}(\Phi^S, P_i^N)$, and P_k^S is the minimum cost scheduling plan for the sink in the set $\text{comp}(\Phi^S, P_j^N)$. Then, $c_i^N \leq c_j^N$ iff $c_h^S \leq c_k^S$. Similarly, let P_h^S be the maximum quality level scheduling plan for the sink in the set $\text{comp}(\Phi^S, P_i^N)$, and P_k^S be the maximum quality level scheduling plan for the sink in the set $\text{comp}(\Phi^S, P_j^N)$. Then, $q_i^N \leq q_j^N$ iff $q_h^S \leq q_k^S$.*

Our algorithm progressively achieves a (sub)optimal solution in several iterations. It starts assigning to all the slots of the sensor the most efficient scheduling plan, and by finding, among the compatible schedules of the sink, the most efficient one. After this preliminary step, if the solution is admissible, then the algorithm enters a loop in which it tries to improve it. The improvement consists in increasing the overall QoS by considering only the scheduling plans of the sensor that have a better quality level than that of the scheduling plan used for the initial assignment. It selects among these the most efficient plan and tries to reassign slot by slot (starting from the slot at sunrise, in which the solar cells start producing energy again) the scheduling plan for the sensor and, for each reassignment, it also finds a (new) compatible scheduling plan for the sink that, according to Assumption 1, must have a better quality level, thus this reassignment improves the overall quality level of the solution. This loop ends only when there are no more better scheduling plans to try. Otherwise, if the solution is not admissible (it is not energy neutral or the battery level of the device drops below the minimum), the algorithm enters a loop in which it looks for solutions that cost less but that may have also a lower QoS. Only the scheduling plans of the sensor that have a smaller cost than that of the scheduling plan used for the initial assignment are considered. It

selects among these the most efficient and tries to reassign slot by slot (starting from the slot at sunset, in which the solar cells stop producing energy) the scheduling plan for the sensor and, for each reassignment, it also finds a (new) compatible, scheduling plan for the sink that, according to Assumption 1, must have a lower cost, reducing thus the overall cost of the solution. This loops ends only when there are no more scheduling plans with a lower cost to try.

5. EVALUATION

We evaluated our approach by simulation. We select the plans for the sensor and for the sink that maximize the overall QoS according to our algorithm in a time frame of 24 hours divided in $\sigma = 300$ slots. To compute the energy production from the solar cell we take the average monthly solar irradiance in Madrid [5]. We compute the instantaneous energy production at any hour by using the model in Section 3 and then we distribute it equally among the slots in an hour ($\frac{\sigma}{24}$). B_{max}^X is 2500mAh, B_{min}^X is 1950mAh and B_0^X is 2000mAh. Table 1 shows the details for the scheduling plans of the sensor $N1, \dots, N6$ and for the sink $S1, \dots, S8$. The compatibility between these plans is presented in Table 2, which shows a compatibility of at most a 40% and 80% between plans. We proceed with the assignment algorithm for the 1st day of each month starting at 0:00am that corresponds to slot 0.

	S1		S2		S3		S4		S5		S6		S7		S8	
%	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
N1	0	1	0	1	1	1	0	1	0	0	1	1	0	1	1	0
N2	1	1	0	1	1	1	0	1	1	1	0	1	0	0	0	0
N3	0	1	1	1	1	0	0	1	1	1	0	0	0	1	0	1
N4	1	1	0	1	1	1	1	0	0	0	0	1	0	1	0	1
N5	0	0	0	1	1	1	1	1	0	0	1	1	0	1	0	0
N6	0	1	0	1	0	0	1	1	0	0	1	1	0	1	1	1

Table 2: Compatibility matrix. Legend: L: Low compatibility-40%; H: High compatibility-80%.

We obtain an overall energy neutral system for the compatibilities considered and for all months of the year, which means that the algorithm finds an assignment of scheduling plans to slots in the sensor and in the sink that verifies $B^X(300) > B^X(0)$. Figure 1 shows the battery level for the sensor for a compatibility of 40%. There is a clear relationship between the energy production and the battery level: the bigger is the production, the larger is the battery level achieved at the last light hour (close slot at the sunset r_2) and the lower is the battery level before the slot at the sunrise r_1 , due to the fact that the most consuming plans can be

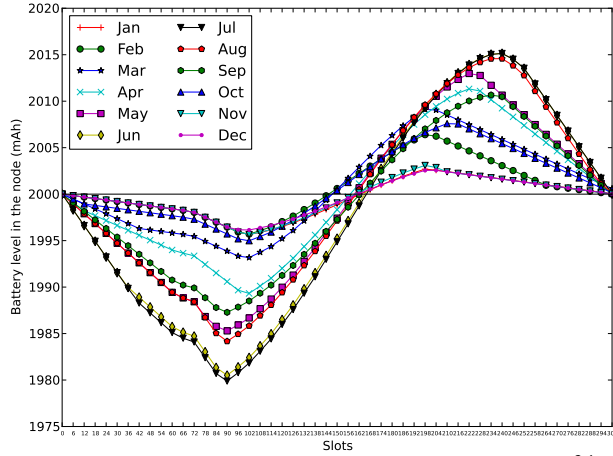


Figure 1: Sensor battery with compatibility=40%.

assigned when the production is larger. The largest battery level is achieved in July at r_2 . Correspondingly, the lowest battery level is achieved at r_1 in the same month in the sensor (the most consuming plans are assigned also in that month). The same experiment was done for the sink, and we observe that the battery level at slot 300 in the sink is rather larger than in the sensor, which means that a part of the energy production is accumulated in the battery (while it is lower than B_{max}) but it is not used to execute a plan with higher QoS.

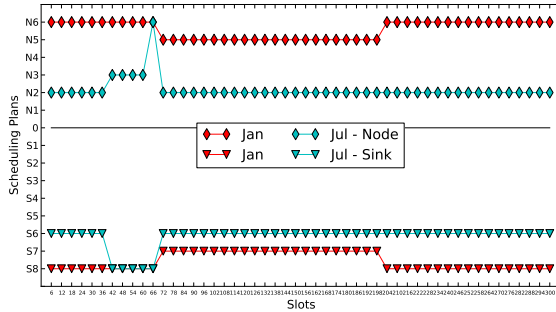


Figure 2: Assignment with compatibility of 80%.

Figure 2 shows the assignment for January and July assuming at most 80% of compatibility. The axis y represents the plans in the sensor and in the sink and the axis x represents the slot where assigned. The order of the plans by efficiency is $N6, \dots, N1$ and $S8, \dots, S1$. In January, as observed, the algorithm starts assigning the most efficient plan in the sensor $N6$ and in the sink $S8$. When the first change occurs in the sensor at slot 72 and the plan $N5$ is selected, the sink looks for the first plan compatible with the new one. Since $S8$ is not compatible with $N5$, $S7$ is selected and starts to execute at the same slot. When the sensor selects again $N6$ at slot 204, the sink comes back to $S8$. We repeat the same experiments with a compatibility of at most 40%. A lower compatibility assigns plans that are farther from $S8$, which increase the overall QoS of the assignment. QoS is computed as the average quality level of the plans assigned and thus is higher in summer than in winter. As expected, the QoS in the sensor is larger than in the sink. Remind that the sink executes the first most efficient plan while compati-

ble with the plan in the sensor, which leads to have a biggest production while the consumption is kept. If we reduce the compatibility we reduce, in turn, the probability of that a plan in the sensor be compatible with the most efficient plan in the sink. Note that the overall QoS with a 40% of compatible plans is higher than with a 80% and that in the months of bigger production the QoS achieved in the sensor is similar to the sink; this is because the sensor upgrades several times to achieve the plan with the highest quality able to execute and, at the same time, with a low compatibility we reduce the possibility of that such a plan be compatible with the plan of the sink, forcing to the sink to select another plan that is less efficient but has a higher consumption and a higher quality, which improves the QoS in the sink. However, with a higher compatibility, the QoS achieved in months of bigger production could be lower than in months of lower production; e.g. in April, May, and September the QoS in the sink is close to 50%, which indicates that $S8$ is compatible with most of the plans selected in the sensor.

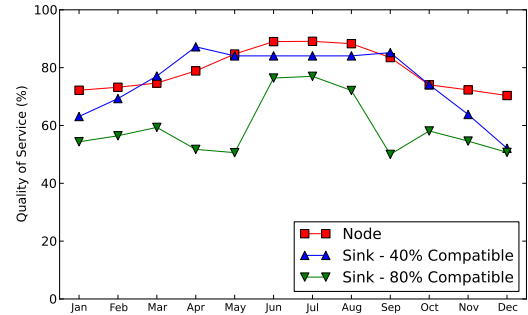


Figure 3: QoS for the sensor and for the sink.

6. FUTURE WORK

We address the problem of the energy management concerning to two networked devices, a sensor and a sink, equipped with solar cells, with the aim of achieving potentially infinite lifetimes. Our algorithm achieves a high QoS for the sensor and preserves a low cost in the sink, leading to accumulate the energy produced but not consumed into the battery, which could be further used to execute another tasks or attending a larger number of sensors. Our next step is to generalize our algorithm to consider N nodes.

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